



SOC Final Spring 24 (Solved)

Society, Environment and Engineering Ethics (United International University)



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SOC

Final Exam Spring 2024

1. GreenTech Innovations is a well-known software company that specializes in developing advanced environmental monitoring systems. The company is reputed for its employee-centric policies, providing excellent work-life balance, and regularly conducting training programs to keep staff updated with the latest state-of-the-art technologies. GreenTech is also distinguished for its exceptional client support. The company not only delivers cutting-edge software but also prepares detailed documentation and usage guides to further enhance client experience. Recently, a rural tribal community approached GreenTech to develop a new environmental monitoring system after their previous system became outdated and unsupported. This system was vital for the community, as they depended on it for sustainable agriculture and fishing. However, during the development of this software, GreenTech significantly deviated from their standard practice of closely collaborating with clients to customize solutions, assuming the tribal users had limited software knowledge. The company engaged minimally with the tribal community while developing the initial version. Furthermore, several bugs identified by the community were overlooked as minor issues, despite their potential impact on the community's essential activities. With a view to quickly deploy the new system, GreenTech accessed historical data from the old system without obtaining explicit authorization from the tribal leaders. Moreover, the old system was deactivated without ensuring that all user data had been successfully migrated to the new system. Furthermore, GreenTech reassigned a skilled front-end developer, who typically focused on UI functionalities, to oversee AI development for the new platform. Due to the sudden nature of this task switch and the nearing of deadlines, the developer struggled to meet the specific needs of the tribal community. Analyze the scenario described above using the ACM Code of Ethics, stating with reason which of the principles are being followed and which are being violated.

Ans: GreenTech Innovations is known for its employee-friendly environment and great client support, but while developing a new environmental monitoring system for a rural tribal community, the company deviated from its usual standards. They failed to collaborate closely with the community, overlooked bugs in the system, accessed old data without permission, and reassigned developers to tasks they were unfamiliar with, leading to issues in the software.

Ethical Analysis Using ACM Code of Ethics:

i) Principles Followed:

- **1.1 Contribute to society and human well-being:** Greentech's goal of providing a critical system to support sustainable agriculture for the tribal community aligns with this principle.

ii) Principles Violated:

- **1.2 Avoid harm:** By ignoring bugs and rushing the deployment, GreenTech caused potential harm to the community's essential activities.
- **1.3 Be honest and trustworthy:** GreenTech accessed historical data without permission and failed to ensure the proper migration of user data, breaching trust.
- **2.5 Give comprehensive and thorough evaluations:** The company neglected proper feedback from the community and ignored crucial bugs, showing a lack of thoroughness.
- **2.2 Maintain high standards of professional competence:** Assigning a front-end developer to an unfamiliar AI role resulted in a subpar product, violating this principle.

In conclusion, while GreenTech aimed to help the community, several key ethical principles were violated in the process, particularly around trust, harm prevention, and professional competence.

2. Imagine a scenario where a university like UIU, provides interest-free loans to students during times of financial need. Currently, a team manually assesses the eligibility of each student for the loan, which can be time-consuming. The team is considering developing a machine learning-based tool to speed up the initial screening process. The tool will utilize various machine learning models to analyze students' academic performance, financial history, and personal circumstances to predict their likelihood of needing financial assistance. Reflecting on this situation, consider the potential ethical implications of implementing such a tool. Discuss concerns related to fairness, transparency, and algorithmic bias in determining loan eligibility. Discuss how relying on machine learning algorithms could impact students' access to financial assistance. Additionally, propose strategies to ensure that the machine learning tool operates ethically, respects students' privacy and dignity, and effectively supports those in need of financial assistance.

Ans: University is considering using a machine learning tool to assess students' eligibility for interest-free loans. This tool will analyze academic performance, financial history, and personal details to predict which students may need financial assistance. While this system could speed up the process, there are ethical concerns about fairness, transparency, and bias.

Ethical Concerns:

- i) **Fairness and Bias:** Machine learning algorithms can unintentionally reinforce biases. For example, if historical data is biased (favoring certain demographics), the model might unfairly exclude deserving students from financial help.
- ii) **Transparency:** The decision-making process of machine learning tools can be opaque, making it hard for students to understand why they were accepted or denied, potentially leading to mistrust.
- iii) **Impact on Access to Financial Assistance:** Relying heavily on algorithms could result in some students, particularly those from marginalized backgrounds, being unfairly screened out if the model is not properly designed or tested.

Strategies to Ensure Ethical Use:

- **Ensure Fair Data:** Use diverse and unbiased datasets, regularly checking for and correcting any algorithmic bias.
- **Provide Transparency:** Clearly explain how the tool works and allow students to appeal decisions.
- **Protect Privacy:** Respect students' personal data and implement strong privacy safeguards.
- **Human Oversight:** Use the tool only for initial screening and keep humans involved in the final decision-making to ensure fairness.

3. In a project named “EmoBoost”, TikTok collaborated with AI experts and psychologists to analyze the behavior of users who frequently engage with content related to mental health challenges, such as anxiety and stress. TikTok quietly monitored the interactions and content consumption patterns of 5,000 users who expressed struggles with mental well-being. Using advanced algorithms, the platform identified the types of content that alleviate stress and promote well-being for these users. Building on this research, TikTok introduced a new “MoodLift” feature in December 2023. Whenever the AI detected a user engaging with content related to mental health challenges, it would intervene by suggesting uplifting videos, meditation exercises, or mental health resources tailored to the user’s preferences. The initial feedback from users was overwhelmingly positive, with many reporting a sense of relief and gratitude for the supportive content provided by TikTok. In a press release, TikTok highlighted the success of the “EmoBoost” project in enhancing user experience and promoting mental well-being on the platform. Reflecting on TikTok’s “EmoBoost” project, what ethical considerations emerge regarding the platform’s monitoring of users’ mental health-related content and the implementation of AI-driven interventions? How might Tik Tok ensure transparency, informed consent, and user autonomy while researching and deploying features to support users’ mental well-being?

Ans: In a project called "EmoBoost," TikTok worked with AI experts and psychologists to monitor 5,000 users who engaged with mental health-related content, like anxiety and stress. Using AI, TikTok found out what type of content helped users feel better and introduced a feature called "MoodLift" in December 2023. This feature suggested uplifting videos and mental health resources whenever users interacted with such content. Feedback was positive, but there are ethical concerns about TikTok monitoring users and using AI interventions without clear consent.

Ethical Considerations:

- i) **Privacy:** Monitoring users’ interactions with mental health content without their knowledge raises serious privacy concerns.
- ii) **Consent:** Users should be informed and give clear permission if their mental health data is being analyzed and used for interventions.
- iii) **Transparency:** TikTok needs to be open about how the AI works and why certain content is suggested.

Ensuring Ethical Use:

- **Informed Consent:** TikTok should tell users about the monitoring and get their explicit consent.
- **User Autonomy:** Users should have control over whether they want the AI-driven "MoodLift" feature or not.
- **Clear Communication:** Explain how TikTok uses the data to help users and offer opt-out options.

By being transparent and respecting user privacy, TikTok can ethically support mental well-being.

4. In a small town, there is a new gym where people love to work out. They think it is just a regular place to get fit, but something secret is going on. The gym owners are actually scientists secretly studying how exercise and special drinks affect the human body. They're trying to make people stronger, like a super army! When people lift weights or run on treadmills at the gym, sensors hidden in the equipment track how hard they are working. And when they drink the special drinks provided by the gym, the scientists watch how it changes their energy levels. But there is a catch. The special drinks might have side effects. Some people feel jittery or have trouble sleeping after drinking them. Others notice strange changes in their mood or appetite. The scientists are keeping a close eye on these side effects, but they are determined to keep their experiments going. All the while, the people working out have no idea they are part of a big experiment. They are just trying to get healthier, but little do they know, they might be helping create super-powered soldiers with some unexpected consequences. Is the data collection process correct? What necessary steps should be taken? Explain.

Ans: In a small town, people love going to a new gym, but they don't know something secret is happening. The gym's owners are scientists studying how exercise and special drinks affect the body, hoping to make people stronger. They secretly track workout performance using hidden sensors and watch how the drinks change energy levels. However, some people feel jittery, can't sleep, or notice changes in mood after drinking the gym's special beverages. The scientists keep experimenting, but gym-goers have no idea they're part of this study.

Is the data collection process correct?

- **No, it is not ethical.** The gym-goers are unaware they are part of an experiment, which violates their right to **informed consent**. They should be informed and asked for permission before any data is collected or analyzed.

Necessary Steps to Take:

- i) **Get Consent:** Inform participants about the study and get their permission to collect data.
- ii) **Disclose Side Effects:** Warn people about potential side effects of the special drinks.
- iii) **Ensure Privacy:** Make sure personal data is kept safe and secure, with clear privacy policies.
- iv) **Offer Opt-Out:** Give people the option to not participate in the study if they don't want to.

By following these steps, the scientists can run their experiments ethically and respect the rights of the gym-goers.

5. In 2024, a consortium of technology companies, urban planners, and environmentalists launched a "Smart City" initiative in a major metropolitan area. The initiative aims to integrate advanced IoT devices, AI-driven analytics, and sustainable energy solutions to enhance urban living. Key features of the smart city include autonomous public transport, energy-efficient buildings, and a comprehensive waste management system that uses AI to optimize recycling processes. While the project promises significant improvements in energy efficiency and reduced carbon footprints, concerns have been raised regarding the environmental costs of implementing such technologies. Critics argue that these high-tech systems' production, maintenance, and eventual disposal could offset the potential environmental benefits. They point to the extensive use of rare earth metals, the increased electronic waste from outdated systems, and the potential for these technologies to contribute to a larger digital divide in urban areas. Analyze the environmental impacts of these Smart City technologies, both positive and negative, considering their entire lifecycle. Propose strategies to reduce negative impacts and enhance positive effects.

Ans: In 2024, a group of tech companies, urban planners, and environmentalists launched a "Smart City" initiative in a large city. The project uses advanced technology like AI, IoT devices, and sustainable energy solutions to improve city life. Key features include self-driving public transport, energy-efficient buildings, and AI-powered waste management to boost recycling. Although this project promises better energy use and a smaller carbon footprint, some people worry that the environmental costs, like the production and disposal of technology, may cancel out the benefits.

Environmental Impacts:

i) Positive Impacts:

- **Energy Efficiency:** AI systems and smart devices help reduce energy waste in buildings and transportation.
- **Reduced Carbon Footprint:** By using sustainable energy sources and optimizing waste management, cities can lower their overall carbon emissions.

ii) Negative Impacts:

- **Use of Rare Earth Metals:** Many high-tech devices require rare earth metals, which are environmentally damaging to mine.
- **Electronic Waste:** As technology becomes outdated, disposing of old systems can create a lot of e-waste.
- **Digital Divide:** High-tech solutions may not be equally accessible to all citizens, leading to inequality in access to smart city benefits.

Strategies to Reduce Negative Impacts:

- i) Eco-Friendly Tech:** Invest in developing tech that uses fewer rare earth metals and is more energy-efficient.
- ii) Recycling Programs:** Implement strong e-waste recycling systems to manage outdated devices.
- iii) Inclusive Planning:** Ensure that smart city technology is accessible to all, reducing the digital divide.

By focusing on these strategies, the Smart City initiative can maximize its positive environmental impacts while minimizing the negative ones.